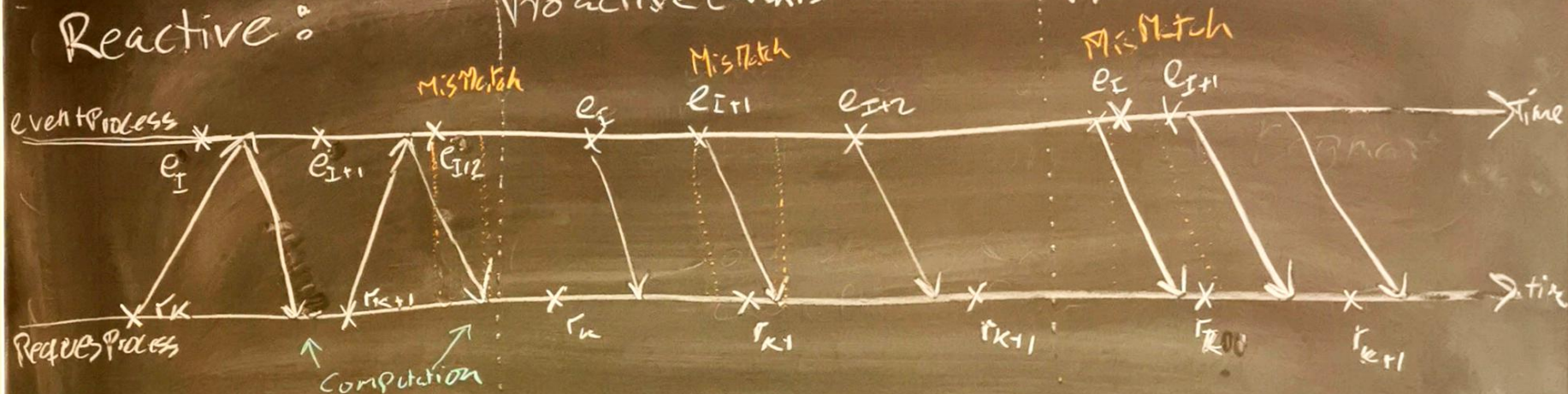


6 Performance implications of data access schemes

Reactive:

Proactive (event)

Proactive (periodic)



Scenario Problem: Stale data $\rightarrow$ Controller bad decision

Key Metric: mmPr (Mismatch Prob.)

$$\text{mmPr}(\epsilon) := \Pr(|X_s(c_i) - X_c(c_i)| > \epsilon)$$

threshold

true value

controller knowledge at time c_i

Analysis Results: (all assume Poisson event process rate λ)
Downstream delay D rate $\checkmark$

Reactive:

- Mismatch if sensor value change during round trip

$$\text{mmPr} = 1 - \mathcal{L}\{fD\}(\lambda) \xrightarrow[\text{if eq. delay}]{\lambda/(\lambda+\nu)} \lambda/(\lambda+\nu) \quad \text{if } \nu \gg \lambda \text{ low mm}$$

Proactive event full updates:

- MM if last MSG in transit

- Same mmPr as reactive

because same question: did most recent msg arrive